

Vectors and Geometry of Space

Vector: A vector is a quantity having both magnitude and direction.

In 2D vector is written as $\vec{a} = a_1\hat{i} + a_2\hat{j}$

Or, $\vec{a} = (a_1, a_2)$

In 3D vector is written as $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

Or, $\vec{a} = (a_1, a_2, a_3)$

Where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the x, y and z-axes.

Magnitude: Magnitude of a vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ is represented by

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Vector Operations:

1. Addition.
2. Subtraction.
3. Multiplication.

Distance between two points: Distance between two points $P(x_1, y_1, z_1), Q(x_2, y_2, z_2)$

$$PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Dot Product: Dot product for two vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Also

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

Where θ is the angle between the vectors.

Properties:

1. Commutative

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

2. Distributive

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

Orthogonality

Two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if $\mathbf{a} \cdot \mathbf{b} = 0$

Cross Product: Cross product for two vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Magnitude: $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\Theta$

Properties: $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$

$\mathbf{a} \times \mathbf{a} = \mathbf{0}$

➤ Two vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$

Equation of a line: Equation of a line in 3D passing through a point $A(x, y, z)$ and parallel to vector $\mathbf{b}(a, b, c)$.

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}$$

Example-1: For which value of a , the vectors $a\hat{i} - 2\hat{j} + \hat{k}$ and $2a\hat{i} - a\hat{j} - 4\hat{k}$ are perpendicular to each other.

Solution: The given vectors will be perpendicular to each other if their dot product is equal to zero.

$$\text{Therefore, } (a\hat{i} - 2\hat{j} + \hat{k}) \cdot (2a\hat{i} - a\hat{j} - 4\hat{k}) = 0$$

$$2a^2 + 2a - 4 = 0$$

$$a^2 + a - 2 = 0$$

$$a^2 + 2a - a - 2 = 0$$

$$a(a + 2) - 1(a + 2) = 0$$

$$(a + 2)(a - 1) = 0$$

$$a = -2, 1$$

If $a = -2, 1$ then the vectors will be perpendicular.

Example-2: For which value of m , the vectors $7\hat{i} - \hat{j} + 3\hat{k}$ and $14\hat{i} - 4m\hat{j} + 6\hat{k}$ are parallel to each other.

Solution: The given vectors will be parallel to each other if their cross product is equal to zero.

Therefore,

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -1 & 3 \\ 14 & -4m & 6 \end{vmatrix} = 0$$

$$\hat{i}(-6 + 12m) - \hat{j}(42 - 42) + \hat{k}(-28m + 14) = 0$$

Equating the coefficient from both sides

$$-6 + 12m = 0$$

$$m = 1/2$$

Example-3: Show that the vector $\hat{i} + \hat{j} + \hat{k}$, create equal angle with the axes of coordinate.

Solution: Let $\mathbf{r} = \hat{i} + \hat{j} + \hat{k}$

$A = \hat{i}$ (vector along the x-axis)

$B = \hat{j}$ (vector along the y-axis)

$C = \hat{k}$ (vector along the z-axis)

$$|\mathbf{r}| = \sqrt{1 + 1 + 1} = \sqrt{3}$$

$$|A| = |B| = |C| = 1$$

Let θ_1 be the angle between A and $\mathbf{r}$, θ_2 be the angle between B and $\mathbf{r}$, θ_3 be the angle between C and $\mathbf{r}$

$$A \cdot \mathbf{r} = |A||\mathbf{r}|\cos\theta_1$$

$$1 = \sqrt{3}\cos\theta_1$$

$$\cos\theta_1 = 1/\sqrt{3}$$

$$\theta_1 = \cos^{-1}(1/\sqrt{3})$$

Similarly,

$$\theta_2 = \cos^{-1}(1/\sqrt{3}) = \theta_3$$

Therefore the vector creates equal angle with the axes of coordinate.

Example-4: Find the equation of the line passing through the point A (2, -1, 3), and parallel to the vector $\mathbf{b} = \langle 4, 2, -1 \rangle$

Solution: The position vector of A, $\mathbf{a} = 2\hat{i} - \hat{j} + 3\hat{k}$

The direction vector, $\mathbf{b} = 4\hat{i} + 2\hat{j} - \hat{k}$

The vector equation of line,

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}$$

$$\mathbf{r} = (\widehat{2i} - \hat{j} + \widehat{3k}) + t(\widehat{4i} + 2\hat{j} - \hat{k})$$

$$= (2 + 4t)\hat{i} + (2t - 1)\hat{j} + (3 - t)\hat{k}$$

Exercise: 1. $A = 2\hat{i} + 3\hat{j} - \hat{k}$, $B = \hat{i} + 2\hat{j} - \hat{k}$, then find the angle between the vectors $A + B$ and $A \times B$.

2. Find the angle created by the vector $\hat{i} + 2\hat{j}$, with the axes of coordinate.

3. For which value of m the vectors $2\hat{i} + 3\hat{j} - 5\hat{k}$ and $m\hat{i} - 2\hat{j} - 10\hat{k}$ are perpendicular to each other.